MILITARY SPECIFICATION

RECEIVER - TRANSMITTER, LASER: RT-1021 ( )/VVG-1: 11738821

This specification is approved for use by all Departments and Agencies of the Department of Defense.

WARNING - EYE HAZARD

EXTREME CAUTION MUST BE OBSERVED IN OPERATING AND TESTING THE RECEIVER - TRANSMITTER. IF THE LASER LIGHT BEAM ENTERS THE HUMAN EYE, THE RETINA OF THE EYE CAN BE BURNED. THE HIGH ENERGY DENSITY OF THE BEAM CAN ALSO PRODUCE SUCH DAMAGE AT EXTREME RANGE AND BY REFLECTION FROM ANY SHINY SURFACE SUCH AS GLASS, METAL, AND PAINT. IN VIEW OF THE POSSIBILITY OF PRODUCING SERIOUS EYE DAMAGE, THE RECEIVER-TRANSMITTER SHOULD NOT BE OPERATED UNLESS THE TARGET AREA HAS BEEN EVACUATED.

1. SCOPE

1.1 This specification covers one type of electro-optical unit designated Receiver-Transmitter, Laser: RT-1021( )/VVG-1 part No. 11738821 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical

FSC-1240

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Military

MIL-O-13830	Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-C-14880	" Component Assembly: 10559479 (Transmitter)
MIL-I-45607	Inspection Equipment; Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedures for
MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type
MIL-B-60621	Bonding Optical Assemblies (Glass to Metal)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

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DRAWINGS

U.S. ARMY, Frankford Arsenal

F8565556	Pressure Tester
K10955929	Cupola Vision Block Housing
K11665439	Adapter Plate
F11738821	Receiver-Transmitter, Laser: RT-1021 ()/VVG-1: 11738821

PACKAGING DATA SHEET

11738821	Packaging of Receiver-Transmitter, Laser; RT-1021 ()/VVG-1: 11738821

OTHER PUBLICATIONS

A11745424	Bonding of Optical Assemblies, Process for l

(Copies of spedifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activit y or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The receiver-transmitter, herein referred to as the unit, shall be manufactured in accordance with Drawing F11738821 and drawings pertaining thereto.

3.2 General specification. The unit shall meet the following requirements, where applicable, of MIL-F-13926:

a. Order of precedence

b. Dimensions and tolerance

c. Inorganic protective surface finishes

d. Electrical and electronic assemblies

e. Part identification and marking

f. Workmanship.

3.2.1 Soldering. Soldering shall be in accordance with MIL-S-45743, or MIL-S-46844, except that bare copper wire exposure shall be permitted at clipped ends of component leads.

3.2.2 Optical bonding. Optical bonding shall be in accordance with MIL-B-60621 or document Al 1745424.

3.3 Performance. Unless otherwise specified, the unit shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90°F.

3.3.1 Loads, power, signals, and auxiliary equipment. The unit shall perform as specified herein when the loads and auxiliary equipment of table I are connected, and 'the power and signals of table I are applied as specified.

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TABLE I
LOADS, POWER SIGNALS AND AUXILIARY EQUIPMENT

Item	Condition	Characteristics	Connections
1.	Loads:		Connected between the following pins:
1.1	Impedance	160 ohms(Ω) ±5% in series with 62 pF ±10%	J1-e and J1-f
1.2	Impedance	160Ω ±5% in series with 62 pF ±10%	J1-g and J1-h
2.	Power sources:		
2.1	+24V tank power	+24 ±6 Vdc Ripple: ≤200 mV p-p	J4-2(+), J4-3(+) and J4-4 (-), J4-5(-)
2.2	+24V Q-switch	+24 ±2 Vdc Ripple: ≤200 mV p-p	J2-P(+) and J2-W(-)
2.3	+5V	+5.0 ±0.2 Vdc Ripple: ≤25 mV p-p	J2-S(+) and J2-W(-)
2.4	-5V	-6.0 ±0.2 Vdc Ripple: ≤25 mV p-p	J2-T(-) and J2-W(+)
2.5	+15V	+15.0 ±0.7 Vdc Ripple: ≤25 mV p-p	J2-U(+) and J2-W(-)
2.6	+29V	+29 ±1 Vdc Ripple: ≤25 mV p-p	J2-V(+) and J2-W(-)
2.7	-1600V	-2100 ±50 Vdc Ripple: ≤100 mV p-p Source impedance=480 kilohms (kΩ) ±5 percent	J2-Z(-) and J2-W(+)
2.8	+400V	+400 ±20 Vdc Ripple: ≤100 mV p-p	J2-p(+) and J2-W(-)
2.9	Zero to +15V	Variable Ripple: ≤25 mV p-p	J4-10(+) and J4-9(-)

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TABLE I (Continued)

Item	Condition	Characteristics	Connections
	Signal sources:		Applied as specified herein
	Digital type-A	Logical one: +4. 0" ±1.0 Vdc Source impedance: . 6k0 ±15 percent Logical zero: +0.2 ±.2 Vdc. .Pulse width and PRF as specified herein.	
	Digital type-B	Logical one: +4.0 ± 1. OVdc , Source impedance: , 1500 ±3O percent. " Logical zero +0. ? ±0.2 Vdc. Pulse width and PRF as specifiod herein.	
	Auxiliary equipment		connected as shown on figure 1
	Power supply (PFN charge)	' 10559345 or equivalent	
	Pulse forming 'network (PFN)	.10559361 or equivalent	
	Switch	Normally open	
	Resistor	10k ±5 percent	
	Resistor	1k ±5 percent	
	Resistor	1k ±5 percent	
	Diode	" IN4247 or equivalent "	

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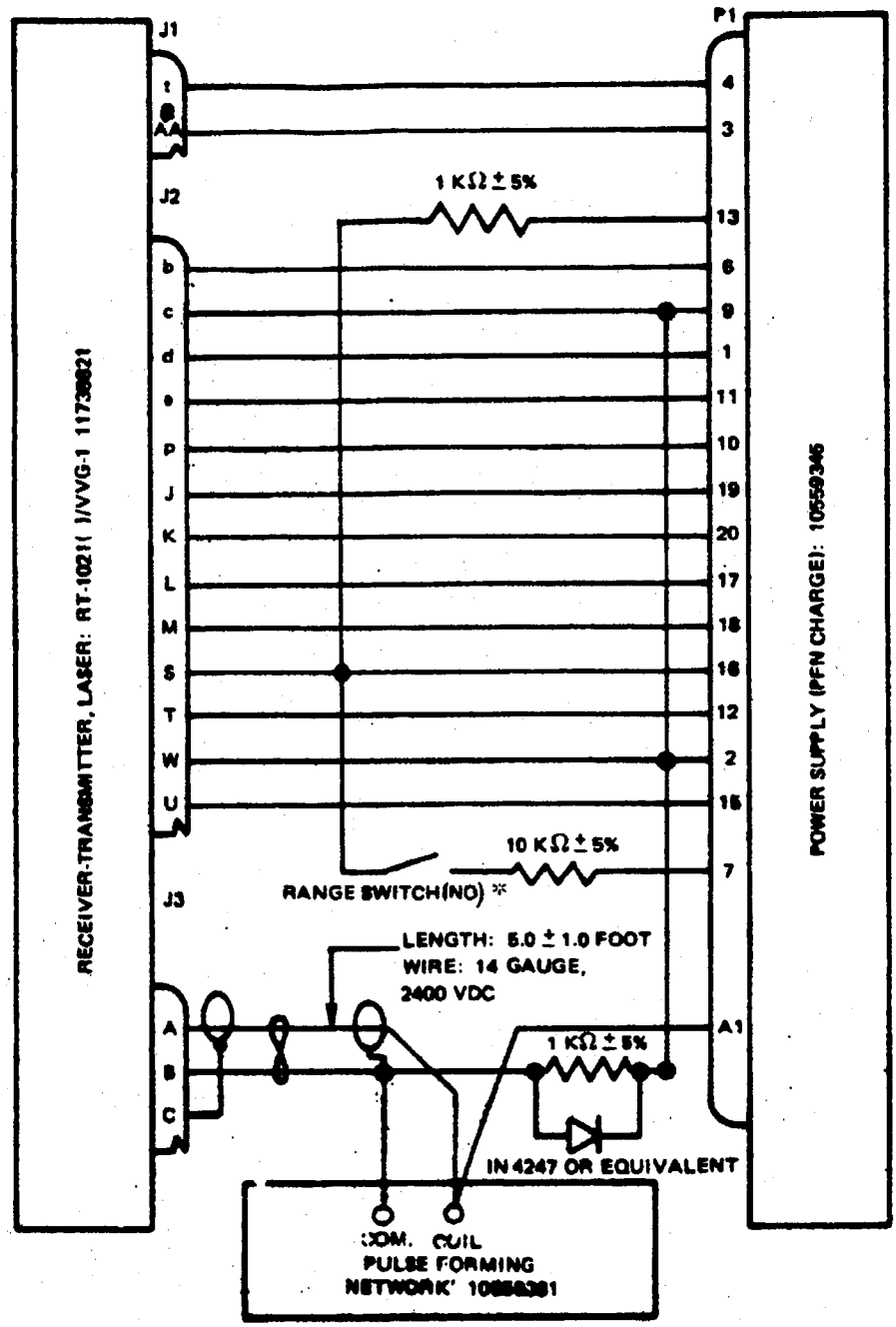


Figure 1. Receiver-Transmitter/Auxiliary Equipment Interconnections

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3.3.1.1 Continuity. Direct continuity (resistance equal to or less than 0.3 ohm) shall exist between pins J1, J2, J3, and J4 as shown on schematic diagram referenced on drawing F11738821.

3.3.1.2 Thermistor resistsance. The resistance between J2-b and J2-c shall be 100k Ω±60 k Ω at room ambient temperature.

3.3.1.3 Lasing pulse timing. During laser firing, the timing between the output at J2-d (SCR Trigger) and the output at J1-h (A-trigger return) shall be as shown on figure 2.

3.3.1.4 A-Trigger, During laser firing, the output at J1-g and J1-h shall be as shown on figure 3.

3.3.1.5 Malfunction" 3 signal. The voltage at J4-35 shall be as follows:

a. +3. 5 ±1.OV after laser firing

b. +0. 2 ±0.2V with the voltage at J2-P removed and a logical zero type-A signal of table I applied momentarily at J2-e.

c. +3.5 ±1.OV with logical zero type-A signal of table I applied momentarily to J4-15

3. 3.1.6 Range switch signal. With the logical zero type -A signal of table 1 applied to, then removed from J4-42, the outputs at J2-h shall be +0.2 ±0.2V- and +4.2 ±1.OV, respectively.

3.3,1.7 Test range 1 nd video signals. With the digital type -B signals (Item 3.2 of table 1) applied to J1-j and. J1-k (waveforms as-shown on figure 4), the outputs at J1-e and J1-f shall be as shown on figure 4. (The signals may be recurring.)

3.3.1.8 Reset signal. After laser firing the waveform shown on figure 5 shall appear at J1-z and J1-AA.

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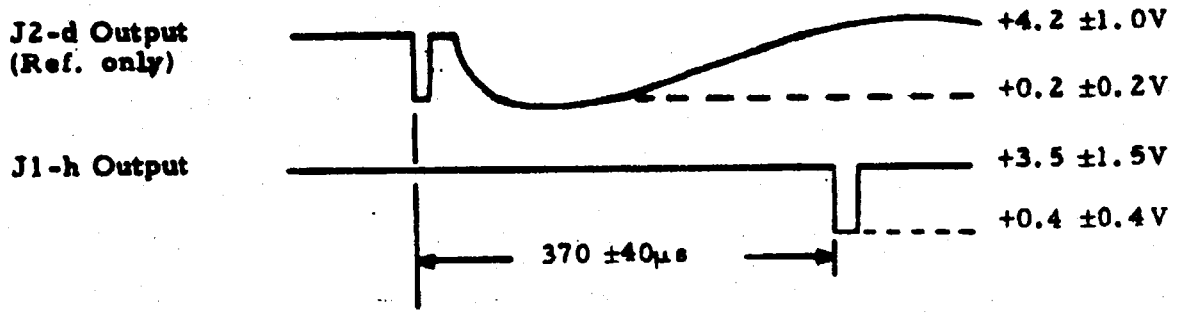


Figure 2. SCR Trigger/A-Trigger Timing Waveforms

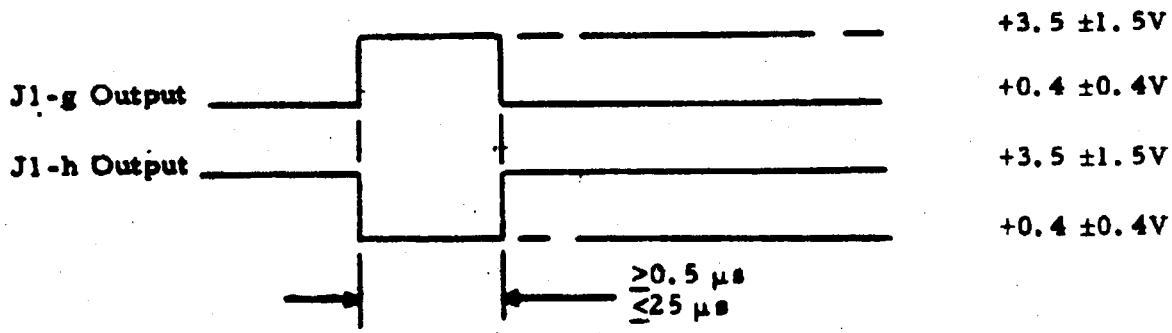


Figure 3. A-trigger Waveforms

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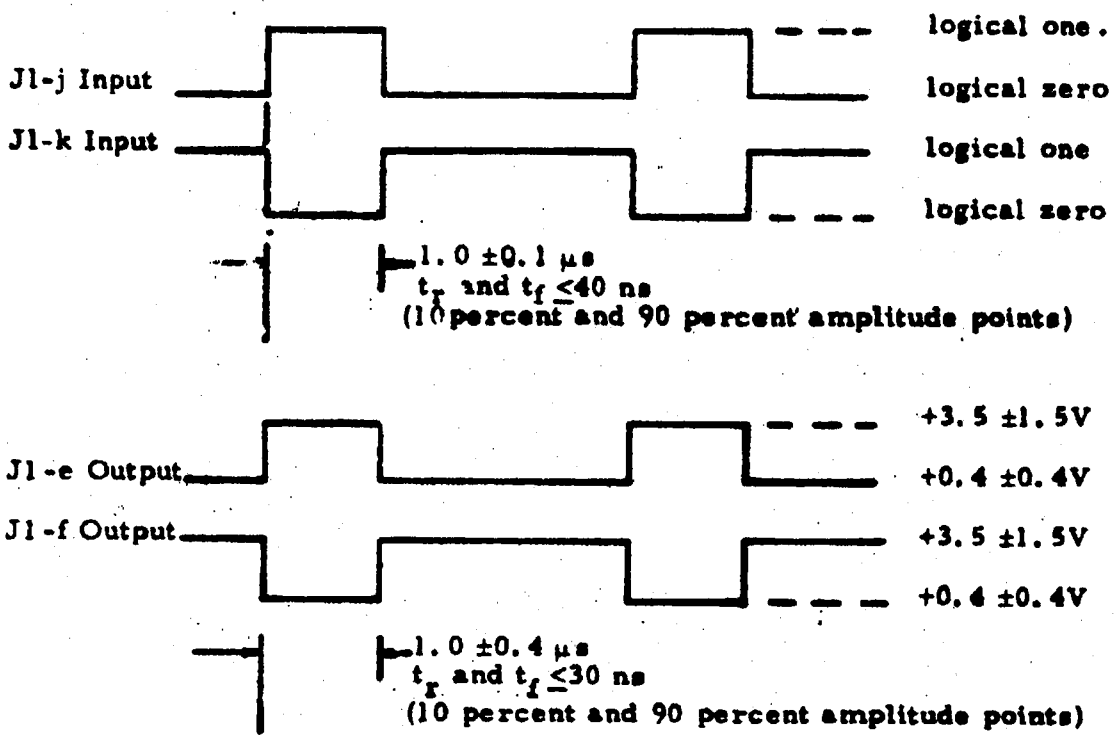


Figure 4. Test Range/Video Waveforms

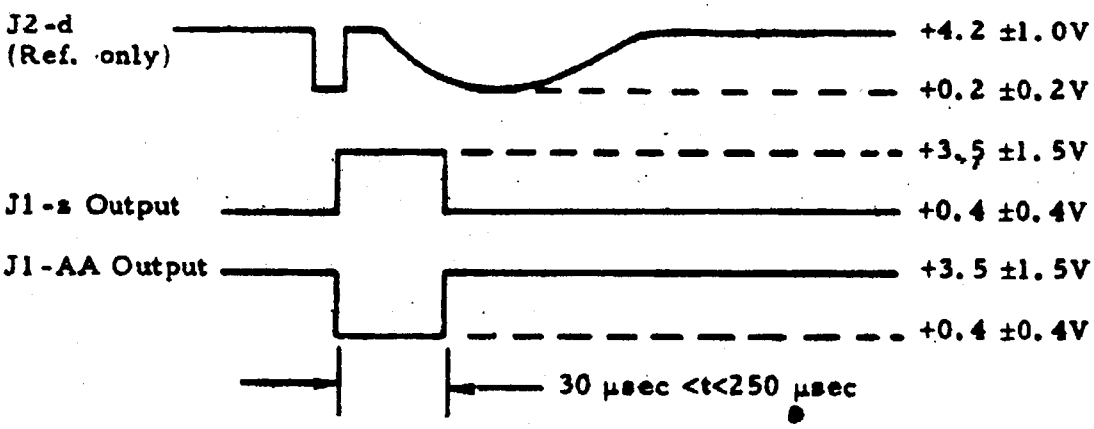


Figure 5. Reset Signal Waveforms

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3,3. 1.9 Maximum ranging. With the input signal to the receiver 1 perture attenuated 37 ±1 decibels (dB), and the laser transitter fired at a diffusely reflecting target located 490 meters from the unit, the output at J1-e shall be as shown on figure 6 .

3.3.1.10 Minimum range inhibit and TPG inhibit. With the input signal to the receiver aperture attenuated 37 ±1dB, with the logical zero digital type-A signal (items 3.1 of table I) applied to J2-f. and the unit fired at a diffusely reflecting target located 490 meters from the unit, either the first pulse as shown in figure 6 or neither pulse shall be present at J1-e. Then with the logical zero digital type A signal (items 3.1 of table I) applied to both J2-f and J2-i, and the unit fired at the same target, both pulses as shown in figure 6 shall be present at J1-e.

3.3.1.11 Events counter. The events counter shall record one event for each activation of the Q- switch or flashtube.

3.3.1.12 AGC inhibit. With the input to the receiver aperture attenuated 37 ±1dB, 1 nd the digital type-B signals of table I 1 pplied to J1-j and J1-k (waveform. 1 s shown on figure 7), the output at J2-j shall be +27 ±3V. With the type-B signals removed from J 1- j and J1-k, the output at J2-j shall be 0±1V.

3.3.1.13 Reticle illumination. With the reticle light control voltage (item 2.8, table I) set for normal illumination, the reticle pattern shall be clearly defined in a darkened area. The brightness of a reticle defect shall be not greater than that of the reticle lines. Increasing the reticle light control voltage shall increase the intensity of the reticle illumination.

3.3.1.14 A-trigger threshold. With the energy output of the unit set to 18.0±3.0 millijoules, the voltage level at J4-35 1 fter laser firing shall be +3. 5 ±1.0 volts.

3.3. 1.15 DioPter 1 djustment

3. 3.1.15.1 Diopter adjustment (+75 ±15°F and +125°F). The required torque for rotation of the diopter adjustment for eyepiece focus at +75 ±15°F and +125° F shall fall within the range of 4 to 18 inch-pounds (in-lbs).

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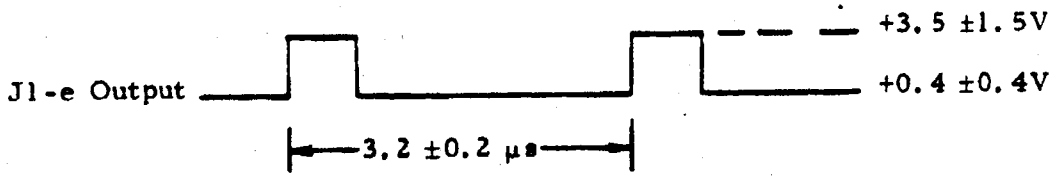
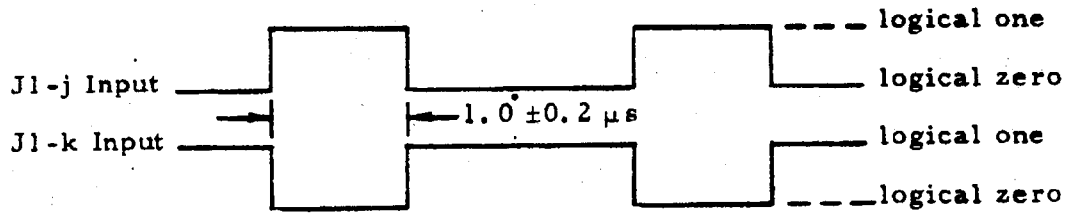


Figure 6. Maximum Ranging Video Signal Waveform



PRF = 1000 ±200 pps

Figure 7. AGC Inhibit Input Signal Waveforms

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3. 3.1.15.2 Diopter adjustment (-40°F). The maximum torque required for rotation of the diopter adjustment for eyepiece focus shall not exceed 20 in-lbs at -40°F.

3.3.1.16 Telescope.

3.3.1.16.1 Cleanliness.

3.3.1.16.1.1 Defects. There shall be no evidence of moisture, grease, fingerprints, condensates,, fractures nor adhesive separations on any glass component when viewing through either end of the sighting system.

3.3. 1.16.1.2 Dirt. There shall be no more than 2 particles of dirt visible in the central area of the reticle within the 5-mil circle. The area of each dirt particle shall not be in excess of the area equivalent to a size 5 dig. There shall be no more than 5 particles of dirt visible in all the other areas of the reticle. The area of each of these dirt particles shall not be in excess of the area equivalent to a size 10 dig. All observations shall be mad. from the eyepiece and, In addition, there shall be no more than 3 particles of dirt or foreign material (classed as dirt) on the entrance window, scanning prism, or eyepiece lens. The size of any dirt particle shall not exceed O. 05 inch.

3. 3.1.16.2 Resolution (sighting system). The resolution on the optical axis shall be six seconds of arc or less. Resolution consists of a-detectable line structure and the proper line count in all four meridians. Focus shift bctween any two meridians shall not exceed 1/4 diopter measured at the telescope eyepiece.

3. 3.1.16.3 Eyepiece focus of reticle. The diopter scale shall indicate zero diopter within plus or minus 1/4 diopter when the eyepiece focus is adjucted to zero diopter with a dioptometer. The range of focus from this position in each direction shall be at least 3 1 /2 diopters as read on the eyepiece seal..

3.3. 1,16.4 parallax (sighting l system). Parallax in the center of the field of view of the sighting telescope shall not exceed 0.1 mil, when viewing a target at a range of 1000 ±100 meters which may be optically simulated,

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3.3.1.17 Sight/transmitter beam alignment. The transmitter beam defined as the center of the 1mil reticle circle, shall be parallel to the receiver line-of-sight within 0. 25 mil.

3.3.1.18 Optical scan angles, The optical scan angles shall be as follows:

a. Elevation: Plus 20 degrees minimum in elevation and minus 10 degrees minimum in depression from a horizontal reference line.

b. Azimuth: 20 degrees minimum total excursion (plus and minus 10 degrees from a vertical reference line).

3.3.1.19 Image tilt. The image of a vertical line shall be vertical within ±2 degrees throughout the elevation excursion range.

3.3.1.20 The target image' observed at the eye-piece shall not shift by more than ±2 degrees throughout the elevation excussion range.

3.3.1.21 Drive train torque and backlash.

3.3.1.21.1 Azimuth. The azimuth drive train startin torque shall not exceed 35 ounce-inch (oz-in) when alternately applying torque in the positive and negative directions to the azimuth positioning shaft of the receiver -transmitter, and when monitoring for movement of the scan prism. Backlash (i.e., the sum of the positive and negative rotation) during the starting torque shall not exceed 30 degrees.

3. 3.1.21.2 Elevation. The elevation drive train starting torque shall not exceed 35 oz-in when applying torque in the positive and negative directions to the elevation positioning shaft of the receiver-transmitter, and when monitoring for movement of the scan prism. Backlash during the starting torque shall not exceed 28 degrees.

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3. 3.2 Environmental

3.3.2.1 storage temperature. The unit shall meet the requirements of 3.3.1.1 through 3.3.1.21.2, except 3, 3.1.15.2, at ambient temperature after exposure to and thermal stabilization at - 80°F and +160°F.

3.3.2.2 Operating temperatures. The unit shall meet the requirements of 3. 3.1.1 through 3.3.1.15 except 3.3.1.2, 3.3.1.9, 3.3.1.10, 3.3.1.12, and 3.3.1.14 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which the requirements of 3. 3.1.1 through 3.3.1.21, except 3.3.1.15,2, shall be met at ambient temperature.

3. 3.2.3 Vibration

3. 3.2..3.1 Vibration "A" The unit shall show no physical failure and shall meet the requirements of 3. 3.1.1 through 3.3.1.21,2, except 3.3.1.15.2, after being exposed to the following vibratory conditions:

a. Double amplitude and acceleration as shown in figure 8

b. Total test time of 150 minutes per axis, except that there shall be no resonance dwells.

NOTE: Vibration "A" shall only be performed when specified in the procurement document (see 6.2e).

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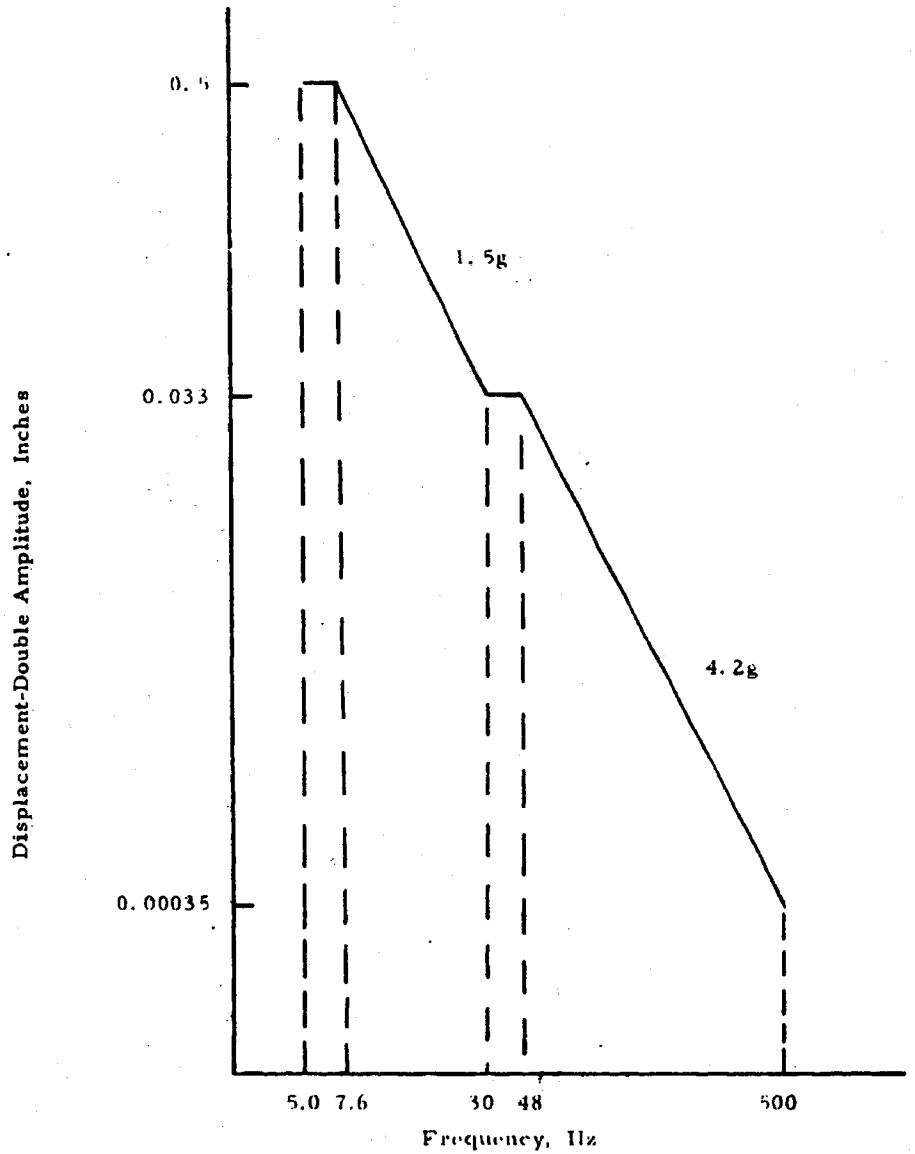


Figure 8. Vibration Envelope

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3.3.2. 3.2 Vibration "B". The unit shall show no physical failure and shall meet the requirements of 3.3. 1.1 through 3.3.1.21.2, except 3.3.1.15.2, after-being exposed to the following vibratory conditions:

a. Amplitude:	1/64 inch, constant (1 /32 inch double amplitude)
b. Vibratory motion:	Simple harmonic
c. Sweep time:	5 to 55 to 5 Hertz in a 5-minute period
d. Vibration cycling:	Three linear or logarithmic sweeps applied to each of the three mutually perpendicular axes (see figure 9).

3. 3.2.4 Shock. The unit shall meet the requirements of 3. 3.1.1 through 3.3.1.21, except 3.3.15.2, after being exposed to a total of 15 shock impulses, three in each direction along the X and Y axes defined on figure 9 and three in the plus (upward) direction along the Z axis. Each shock impulse shall 'be a half-sine wave with a time duration of 3.5 ±0.5 milliseconds and a peak amplitude of 100g.

3.3.3 Sealing. With the unit initially pressurized with nitrogen or air to 5±1 pounds per square inch gage (psig), the loss of pressure shall not exceed 0.055 psig in one-half hour, except for special sampling-environmental during which the loss of pressure shall not exceed 0.22 psig in 2 hours.

3.3.4 Internal pressure. Subsequent to meeting the sealing requirements, the unit shall be dry flushed with an internal atmosphere containing nitrogen having a dew point of at least -63. 5°F. The nitrogen used shall be Type X, Grade B, Class 1 nitrogen conforming to BB-N-411. After the unit is flushed, an internal pressure of 3.5±0.5 psig shall bc established.

3.3.5 Interface The unit shall be capable of interfacing with the Adapter Plate (part no. 11665439) and Vision Block Housing of Cupola (part no. 10955929).

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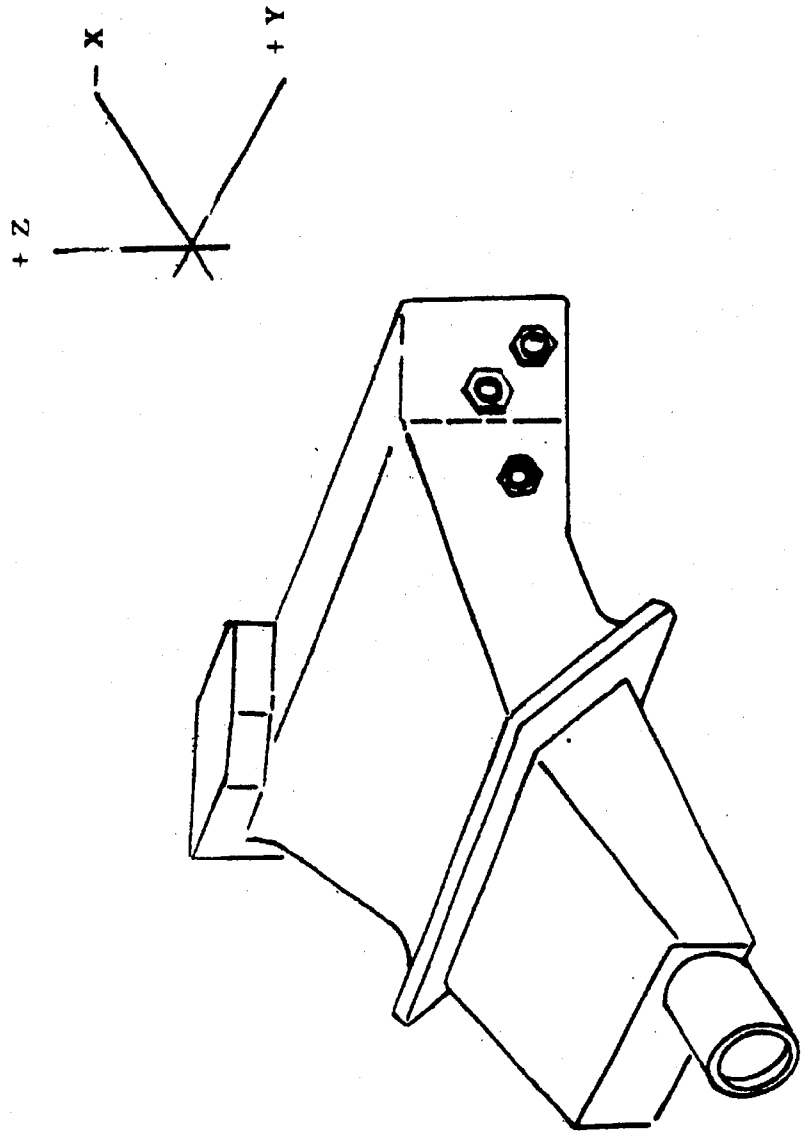


Figure 9. Laser Receiver-Transmitter Axes Definition

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplies is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General Provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three units. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4. 2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification. A record of this inspection, including certificates of conformance for materials. shall be submitted with the sample.

4.202 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified herein or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection and sampling procedures herein and submission of product provisions of MIL-STD-105.

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4.3.2 Examination and tests

4. 3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the respective specifications and Supplementary Quality Assurance Provisions (SQAP)listed in the technical data package (TDP). In the absence of SQAPS, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 First article testing. First article testing shall be conducted in accordance with tables II through IV to determine com - pliance with 4.2. All examinations and tests shall be witnessed by the Government Quality Assurance Representative. When manufacturing processes, materials, or inspection techniques previously applied are subject to change, or an adverse change occurs in performance capability of the unit, this shall be cause to cease those operations contributing to the change or failure until corrections have been made by the contractor and approved by the Government.

4.3.2.3 Functional tests. The requirements and tests in table II shall be inspected on a 100 percent basis. Unless otherwise specified, all examinations and tests shall be conducted at the standard ambient temperature.

4.4 Special sampling The selection of samples shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One unit shall be selected at random by a Government represenative as a control sample from each 50 produced. The control sample shall meet the requirements and tests in table III.

4.4.2 Environmental. Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in table IV.

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

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TABLE II
FUNCTIONAL TESTS

Classi-	Characteristic	Requirement	Test Paragraph
101	Continuity	3.3.1.1	4.6.2
102	Thermistor resistance	3.3.1.2	4.6.3
103	Lasing pulse timing	3.3.1.3	4.6.4
104	A-Trigger	3.3.1.4	4.6.5
105	Malfunction 3 signal	3.3.1.5	4.6.6
106	Range switch signal	3.3.1.6	4.6.7
107	Test range l nd video signals	3.3.1.7	4. 6.8
108	Reset signal	3.3.1.8	4. 6.9
109	Maximum ranging	3.3.1.9	4.6.10
110	Minimum range inhibit and TPG inhibit	3.3.1.10	4.6.11
111	Events counter	3.3.1.11	4.6.12
112	AGC inhibit	3.3.1.12	4.6.13
113	Reticle illumination	3.3.1.13	4.6.14
114	A-trigger threahold	3.3.1.14	4.6.15
115	Diopter adjustment(75 l ±15°F and +125°F)	3. 3.1.15.1	4.6.16.1
116	Diopter adjustment (-40°F)	3.3.1.15.2	4.6.16.2
117	Defects;	3.3.1.16.1.1	4.6.17.1.1
118	Dirt	3.3.1.16.1.2	4.6.17.1.2
119	Resolution (sighting system)	3.3.1.16.2	4.6.17.2
120	Eyepiece focus of reticle	3.3.1.16.3	4.6.17.3
121	Parallax (sighting system)	3.3.1.16.4	4.6.17.4
122	Sight/transmitter beam alignment	3.3.1.17	4.6.18
123	Optical scan angle	3.3.1.18	4.6.19
124	image tilt	3.3.1.19	4.6.20
125	Image wander	3.3.1.20	4.6.21
126	Azimuth	3.3.1.21.1	4.6.22.1
127	Elevation	3.3.1.21.2	4.6.22.2
128	Sealing (1/2 hour)	3.3.3	4.6.24
129	Internal Pressure	3.3.4	4.6.25

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TABLE III
SPECIAL SAMPLING - GENERAL

Classi-fication	Control Test	Requirement	Test Paragraph
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

TABLE IV
SPECIAL SAMPLING - ENVIRONMENTAL

Classi-fication	Control Test	Requirement	Test Paragraph
303.	Storage temperature	3.3.2.1	4.6.23.1
304.	Operating temperature.	3.3.2.2	4.6.23.1
305.	Vibration "A"*	3.3.2.3.1	4.6.23.2.1
306.	Vibration "B"	3.3.2.3.2	4.6.23.2.2
307.	Shock	3.3.2.4	4.6.23.3
308.	Sealing (2 hours)**	3.3.3	4.6.24
309.	Internal Pressure**	3.3.4	4.6.25
310.	Interface	3.3.5	4.6.26

*Performed only when specified in procure document (see 6.2e).
**To be performed after all environmental testing.

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4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table V shall be utilized in performing the tests specified herein.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of the inspection equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAPS forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with test methods and procedures specified in 4.6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of inspection equipment shall be so selected and controlled as to ensure that the inspection equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of inspection equipment.

4.5.2.2.1 Pressure tester. Special test equipment (STE) in accordance with Drawing F8565556 or equivalent, shall be provided to check the sealing of unit.

4.6 Test methods and procedures.

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

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TABLE V
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1	DC Power Supply	+24 ±2 volts at 15 amps Kipple: <200 mv p-p Source impedance: <0.01~
2	DC Power Supply	+5. 0 ±0.2 Vdc Ripple: <25 mV p-p
3	Dc Power Supply	-6.0 ±0.2 Vdc Ripple: <25mV p-p
4	DC Power Supply	+15,0 ±0.7 Vdc Ripple: <25 mV p-p
5	DC Power supply	Zero to +15 Vdc, adjustable Ripple: <25 mV p-p
6	DC Power Supply	+29 ±1 Vdc Ripple: <25 rnv p-p
7	DC Power Supply	-2100 ±50 Vdc Ripple: <100 mV p-p
8	DC Voltmeter	Range: o to 50 volts Accuracy: ±0.3 percent
9	DC Ammeter	Range: O to 100 microampere Accuracy: *1 percent, meter resistance 4400Ø
10	Oscilloscope with Camera	Frequency response to 150 MHz Sensitivity: 5 mV/cm Sweep time: O. 01 psec/cm Camera lens: f/1.3 (Tektronix C40 or equivalent)
11	Thermo Pile*	TRG 108, Tekmar Instruments, Inc. or equivalent
12	Energy Meter	TRG 113, Tekmar Instruments, Inc. or equivalent
13	Signal Source	Capable of providing dc level as specified in table I, item 3.1

*An equivalent for items 11 and/or 12 is model 580 Radiometer system EG&G.

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TABLE V (Continued)

Item	Nomenclature	Description
14	Telescope	Magnification of 3 power (3X) minimum with cross line reticle pattern
15	Target	4 feet square minimum, 80 percent minimum diffuse reflectivity and located at 490 ±15 meters from window of receiver transmitter. The exact target location within this tolerance shall be known within ±1 meter.
16	Optical Test Target	Containing horizontal zero reference. line (indicating vertical reference); a zero azimuthal reference line; ±2 degrees (from vertical) subtense lines (image tilt limits); a pair of near vertical lines, the separation of which subtends 2 degrees in azimuth at the receiver-transmitter (image wander limits): ±10 degrees azimuth (vertical) lines; and -10 degrees, +20 degrees elevation (horizontal) lines.
17	Lens, Long Focal Length	Optical specialties, 105-0829, or equivalent.
18	Traveling Microscope	Unitron TMD (1° x 1°") or equivalent.

4.6.1.1 Initial conditions. Unless otherwise specified herein, connect the loads, auxiliary equipment, power, and input signals to the unit as specified in table I prior to performing the tests specified herein.

4.6.1.2 PFN voltage adjustment. Prior to performing the tests of 4.6.4. initially adjust R10 of the PFN Charge power supply (10559345) for the PFN voltage determined by test and recorded for the transmitter component assembly (10559479), in accordance with MIL-C-14880. Radjust the PFN voltage for a laser energy output of 50 ±5 millijoules.

4.6.1.3 Laser firing Procedure. LAser firing is accomplished by applying a momentary logical zero type A signal to -J (Manual Reset) followed by a momentary closing of the RANGE switch shown on figure 1.

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4.6.2 Continuity test. With all power off, perform resiStance measurements between the pins of J1, J2, J3, and J4 which are directly interconnected. Verify that the requirement of 3. 3.1.1 is met.

4.6.3 Thermistor resistance test. With all power off, measure the resistance between J2-b and J2-c. Verify that the requirement of 3,3.1.2 is met.

4.6.4 Lasing pulse timing test. Perform a laser firing (4.6.1.3) while observing the timing between the output puises at J2-d 1 nd J1-h to determine compliance with 3.3.1.3.

4.6.5 A-Trigger test. Perform a laser firing (4.6.1.3) while observing the outputs at J1-g and J1-h to determine compliance with 3.3.1.4.

4.6.6 Malfunction 3 signal test. The malfunction 3 signal test shall be as follows:

a. perform a laser firing (4.6.1.3) and obserse the output at J4-35 to determine compliance with 3.3.1.5a.

b. Remove the voltage at J2-P. Apply the logical zero type-A signal of table I momentarily to J2-e, and measure the voltage at J4-35 to determine compliance with 3.3.1.5.b.

c. Apply the logical zero type-A signal momentarily to J4-15 and measure the voltage at J4-35 to determine compliance with 3.3.1.5c.

4. 6.7 Range switch signal test. Apply the logical zero type -A signal of table I to J4-42, then remove the signal, and observe the output at J2-h for each input ,condition to determine compliance with 3.3.1.6.

4.6.8 Test range and video signals test. Apply the digital type-B signals of table I to J1-j and J1-k with the waveforms shown -on figure 4, and observe the. output. at J1-e and J1-f to determine compliance with 3.3.1.7.

4. 6.9 Reset signal test. Perform a laser firing (4.6.1.3) while observing the output at J1-z and J1-AA to determine compliance with 3.3.1.8.

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4.6.10 Maximum ranging test. Apply 37 ±1 dB attenuation at the receiver aperture input. Perform a laser firing (4.6.1.3) at the target (item 1 5) specified in table V. Observe the output at J1 -e to determine compliance with 3.3.1.9.

4.6.11 Minimum range inhibit and TPG inhibit test. Attenuate the input signal to the receiver aperture 37 ±1 dB, apply the digital type -A logical zero to J2-f, and fire the unit at the target described in 3.3.1, 10. Verify that the requirement of 3.3.1.10 is met. Then, with the input signal still attenuated 37 ±1 dB, and with the logical zero still applied to J2-f, apply the digital type-A logical zero to to J2-i and again fire the unit at the same target Verify that the requirement of 3.3.1.10. is met.

4.6.12 Events counter test. Observe the performance of the events counter during a laser firing to determine compliance with 3.3.1.11.

4.6.13 AGC inhibit test. Apply 37 ±1 dB attenuation at the receiver aperture input. Apply the digital type-B signals of table I to J1-j and J1-k with the waveforms shown on figure 7. Measure the input at J2-j to determine compliance with 3.3.1.12. Remove the type-B signals from J-1j and J1 -k, and again measure the output at J2-j to determine compliance with 3.3.1.12.

4.6.14 Reticle illumination test. Adjust the reticle light control voltage (item 2.8, table I) to obtain normal illumination. Visually examine the reticle etchings against a darkened background to determine compliance with 3.3.1.13. Vary the reticle light control voltage and observe the intensity of illumination to determine compliance with 3.3.1.13.

4.6.15 A trigger threshold test. Set the energy output level of the unit to 18.0 ±3.0 millijoules. Verify that the requirement of 3.3.1.14 is met.

4.6.16 Diopter adjustment test.

4.6.16.1 Diopter adjustment torque (+75 +15°F and +125°F) test.* This test shall be conducted with standard torque measuring equipment provided with appropriate adapters to accommodate the assembly under test to determine compliance with 3.3.1.15.1. Utilizing an adapte affixed to the diopter knob,measure the running torque to determine compliance with 3.3.1.15.1.

*Dioptcr adjustment torque tests performed at the component level may be utilized to satisfy this test and the test of 4.6.16.2.

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4.6.16.2 Diopter adjustment torque (-40°F) test. Perform the same procedure as specifled in 4.5.16.1 at -40° to determine compliance with 3.3.1.15.2.

4.6.17 Telescope.

4.6.17.1 Cleanliness.

4.6.17.1.1 Defects test. This test shall be conducted by visually examining all visible optical surfaces to determine if the telescope assembly conforms to the requirements of 3.3.1.16.1.1.

4.6.17.1.2 Dirt test. This test shall be conducted by visually examining telescope to determine compliance with requirements of 3.3.1.16.1.2.

4.6.17.2 Resolution (sighting system) test. Perform this test in accordance with the resolution test method and procedure of MIL-O-13830 utilizing an auxiliary telescope of at least 3 power (3x) and the resolving power chart containing four line sets specified therein, to determine compliance with 3.3.1.16.2.

4.6.17.3 Eyepiece focus Of reticle test. The test shall be made with the aid of a calibrated dioptometer with a magnification of at least 3 power (3X) and a diopter range for measuring the accuracies of the diopter knob scale as specified in 3.3.1.16.3. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle and set the dioptometer objective scale to zero. Set the diopter scale of the telescope under test to zero diopter. Position the dioptometer at the telescope eyepiece assembly. While viewing through the dioptometer, obtain best focus of the telescope assembly reticle by adjustment of the dioptometer objective stale.The telescope reticle shall be in best focus within the limit specified in 3.3.1.16.3, as measured on the dioptometer objective scale. Rotate the telescope diopter knob to the clockwise stop, note diopter setting on scale, then rotate the diopter movement counterclockwise to stop. Note diopter setting on diopter scale to determine compliance with the requirement specified in 3 .3.1.16.3 for ran-ge of focus.

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4.6.17.4 Parallax (sighting system) test. Perform this test after focus adjustment has been made in accordance with 3.301.1604. The diopter knob of the telescope shall be set for best focus of the telescope reticle. Parallax requirements shall be met when the target projector collimator is adjusted to the range specified in 3.3.1.16.4. Parallax will be recognized by any apparent displacement of the telescope reticle lines with respect to the target image lines when the observer's head is moved from side to side or up and down. Displacement between the telescope reticle and target image shall not exceed the tolerance specified in 3.3.1.16.4, as measured on the target.

4.6.18 Sight/transmitter beam alignment test. Mount the lens (item 17, table V) in front of the r-t unit window. Locate in the focal plane of the lens apiece of thermally sensitive paper containing target grids. Sight through the eyepiece and position the receiver reticle symmetrically over the juncture of a target grid. Fire the laser. Position the receiver reticle over the juncture of four more target grids and fire the laser at each of these target grids in turn for a total of five firing.Dismount the thermally sensitive paper. Measure with a microscope (item 18, table V) the average deviation of the centers of the five burn spots with respect to the target grid junctures to determine compliance with 3.3.1.17.

4.6.19 Optical scan angles test; Rotate the azimuth and elevation controls on the receiver-transmitter control to the zero position. Place a test target (item 16 of table V) with reference scan angle designation for azimuth (+ 10°) and elevation - 10°, 20°) in front of the rangefinder. Rotate the azimuth control in maximum clockwise and counterclockwise directions and observe the center of the reticle pattern with respect to the reference designation. Rotate the azimuth -control to zero position and repeat the test for elevation control to determine compliance with 3.3.1.18.

4,6.20 Image tilt test. Perform this test utilizing the auxiliary viewing telescope (item 14 of table V) and a calibrated target (item 16 of table V-). Position the target in front of the rangefinder telescope with the vertical reference lines plumb and the zero reference point aligned with the telescope line of sight azimuth and elevation scan control set at zero). Mount the auxiliary telescope with the vertical reticle line plumb at the rangefinder telescope eyepiece assembly, and center the intersect point on the telescope reticle on the target zero reference point. Rotate the elevation scan control over the limits (-10°, +20°) and view the target image through the auxiliary telescope to determine compliance with 3.3.1.19.

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4.6.21 Image wander test. This test shall be conducted in conjuction with the test in 4.6.20. While observing the established position (intersecting point) of the auxiliary telescope reticle, rotate the elevation scan control of the rangefinder as specified in 3.3.1.20. Observe the image wander of the target as a result of elevatioa and depression to determine compliance with 3.3.1.20.

4.6.22 Drive train torque and backlash test.

4.6.22.1 Azimuth test.Apply starting torque to the azimuth positioning shaft the receiver-transmitter control. Monitor the movement of the scan prism (optically or mechanically). Verify that the requirement. bf 3.3.4.21.1. are met.

4.6.22.2 Elevation test. Apply starting torque to the. elevation positioning shaft of the receiver-transmitter control. Monitor the movement of the scan prism (optically or mechanically). Verify that. the requirements of 3. 3. 1.21.2 are met,

4.6.23 Environmental test.

4.6.23.1 Temperature test. The test appaaatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that the unit may be mounted on the inside surface of the test chamber. Perform this test in accordance with Procedure I of MIL-F-13926, with conditions C (-40° F) and E (+ 125° F), and with the rate of temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the tests of 4.6.4 through 4.6.16.2, except 4.6.3, 4.6.10, 4.6.11, 4.6.13, and 4.6.15 to determine compliance with 3.3.2.2. At ambient temperature, perform the tests of 4.6.2 through 4.6.22, except 4.6.14.2, to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.23.2 Vibration test.

4.6.23.2.1 Vibration "A" test. Perform this. test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Procedure VIII therein, except that the vibratory conditions of 3.3. 2.3.1 shall apply. After vibration exposure examine the unit for structural failures, Loose hardware, wire breakage, abrasion and contamination of optical components. Perform the tests of 4.6.3 through 4.6.22, except 4.6.16.2, to determine compliance with 3.3.2.3.1.

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4.6.23 .2.2 Vibration "B". test. Perform this test Utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Part 1 of Procedure I therein, except that the vibratory conditions of 3.3.2.3.2 shall apply. Following vibration exposure, examine the unit for structural failures, loose hardware, wire breakage, and abrasion. Perform the tests of 4.6.3 through 4.6.22, except 4.6.16.2, to determine compliance with 3.3.2.3.2.

4.6.23.3 Shock test. Perform this test utilizing apparatus specified in Method 516.1 of MIL-STD-810, in accordance with Procedure I therein and with the conditions specified in 3.3.2.4. Perform a visual-tactile examination for damage and loose parts. the tests of 4.6.3 through 4.6.22, except 4.6.16.2. to determine compliance with 3.3.2.4.

4.6.24 Sealing test. Perform this test utilizing the special test equipment specified in 4.5.2.2.1. Pressurize the unit as specified in 3.3.3. Measure the pressure loss over the applicable time period to detemine compliance with 3.3.3.

4.6.25 Internal pressure test. Perform this test utilizing the STE specified in 4.5.2.2.1. Subsequent to meeting the sealing requirement, "dry flush" the unit with the nitrogen specifid in 3.3.4 and establish a dry nitrogen atmosphere. Verify that the requirement of 3. 3.4 is met.

4.6.26 Interface test. This test shall be conducted utilizing a limit gage to determine compliance with 3.3.5.

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5. PREPARATION FOR DELIVERY

5.1 Packing, packing. and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11738821. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 intended use. The Laser Receive - Transmitter RT-1021( )/VVG-1 is a unit of the Fire Control Rangefinder AN/VVG-1 ( ). The Laser Receiver-Transmitter contains the laser transmitter, laser receiver, and alignment optics.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Applicable packaging data sheet number (see 5.1)

c. Selection of applicable levels of preservation, packaging and packing

d. Applicable stock number

e. Requirement for First Article Submission which may include 1 dditional environmental tests selected by the procuring agency: eg Vibration "A" (See 3. 3.2.3.1).

Custodian:

Army - Frankford 1

Preparing Activity:

Army - MU

PROJECT NO. 1240-A314

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